1. Task: Simple Task Manager API + UI
Problem:
Build a basic task manager where users can create, update, delete, and mark tasks as completed.
Requirements:
Backend: REST API (CRUD for tasks)
Fields: title, description, status, created_at
Frontend: List + Add + Toggle Complete
Deliverables:
Working UI connected to backend
API endpoints documented (Postman/README)
Basic validation (no empty title)

2. Task: Authentication System (JWT)
Problem:
Implement user authentication system.
Requirements:
Register + Login
Password hashing
JWT-based authentication
Protected route (e.g., /profile)
Deliverables:
Auth flow working end-to-end
Token handling in frontend
Secure password storage

3. Task: Product Listing with Filters
Problem:
Create a product listing page with filtering and sorting.
Requirements:
Backend: Products API
Fields: name, price, category
Filters: price range, category
Sorting: price (asc/desc)
Deliverables:
Filtered UI updates dynamically
Efficient query handling (not brute-force frontend filtering)

4. Task: File Upload System
Problem:
Allow users to upload and view files.
Requirements:
Upload image/file
Store locally or cloud
Display uploaded files
Deliverables:
Upload API
Preview in frontend
Handle invalid file types

5. Task: Real-Time Chat (Basic)
Problem:
Build a simple chat system.
Requirements:
2 users can send messages
Use WebSockets (or polling if they can't handle sockets)
Show message history
Deliverables:
Real-time or near-real-time updates
Clean UI for chat

6. Task: Dashboard with Analytics
Problem:
Build a dashboard showing some metrics.
Requirements:
Backend returns stats (e.g., total users, tasks completed)
Frontend shows charts (Chart.js or similar)
Deliverables:
At least 2 charts
Clean UI layout
Proper API integration

7. Task: Notes App with Search
Problem:
Create a note-taking app with search functionality.
Requirements:
CRUD notes
Search by keyword
Highlight results (optional)
Deliverables:
Fast search (not laggy)
Clean UI
Backend query optimization (basic)

8. Task: Role-Based Access Control
Problem:
Extend an app to support roles.
Requirements:
Roles: admin, user
Admin can delete users/tasks
User has limited access
Deliverables:
Middleware for role checking
UI adapts based on role

9. Task: REST API + Pagination
Problem:
Build an API that returns paginated results.
Requirements:
Endpoint: /items?page=1&limit=10
Return total count + pages
Frontend pagination UI
Deliverables:
Efficient pagination (not load all then slice)
Clean navigation UI

10. Task: Deployment + Docker (Basic)
Problem:
Containerize and run the app.
Requirements:
Dockerfile for backend (and frontend if possible)
Environment variables support
Deliverables:
App runs via docker run or docker-compose
README with setup steps